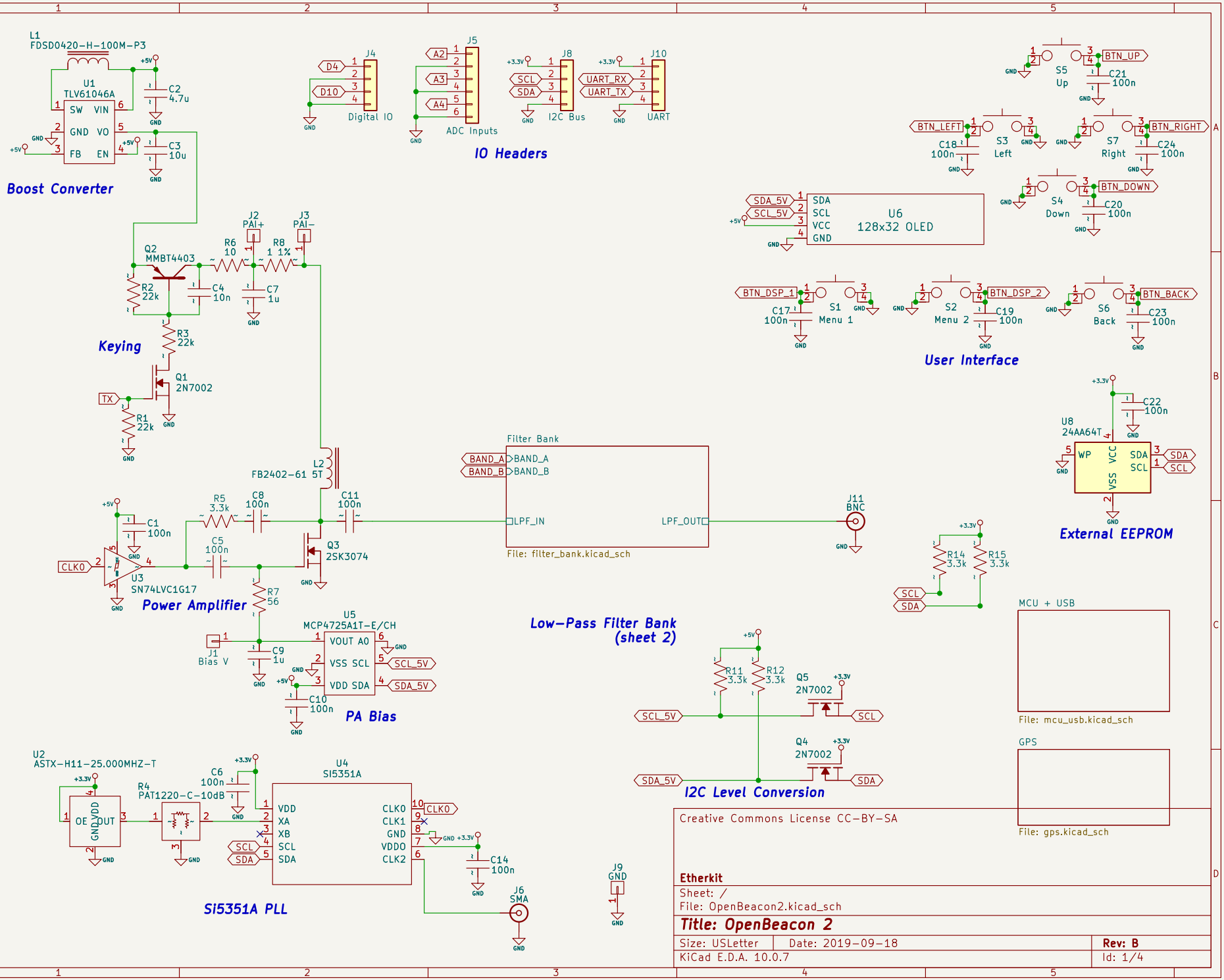
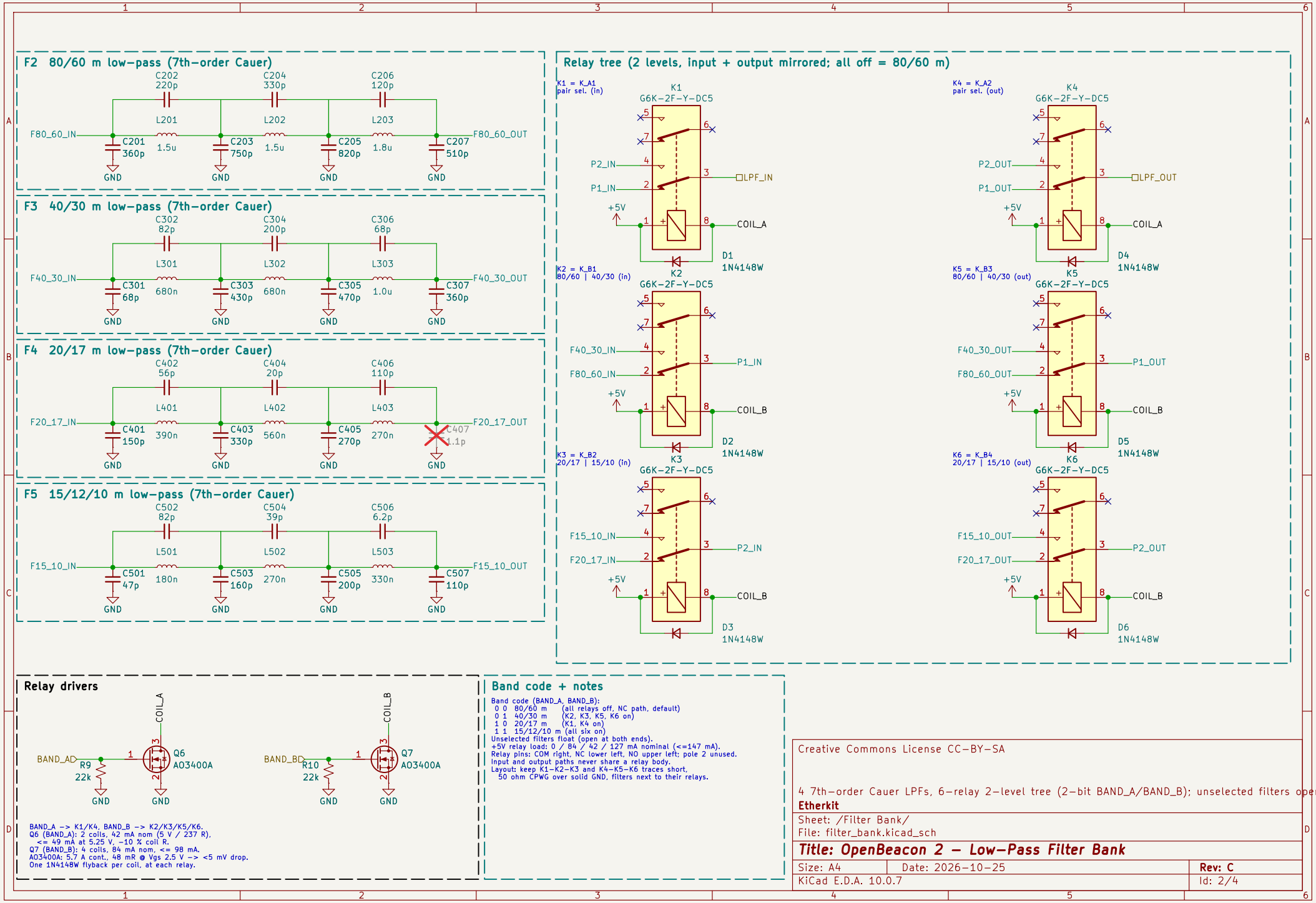
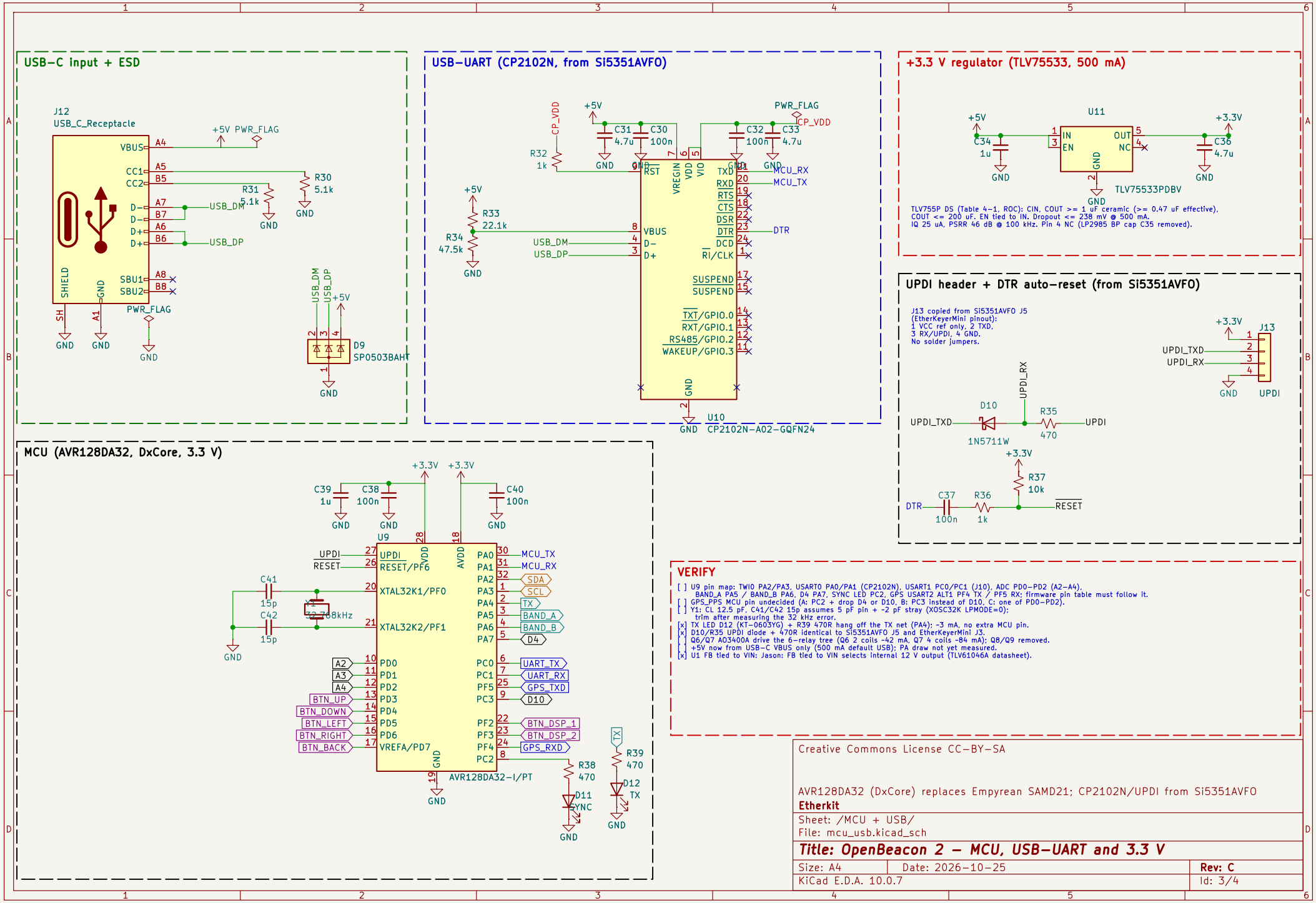
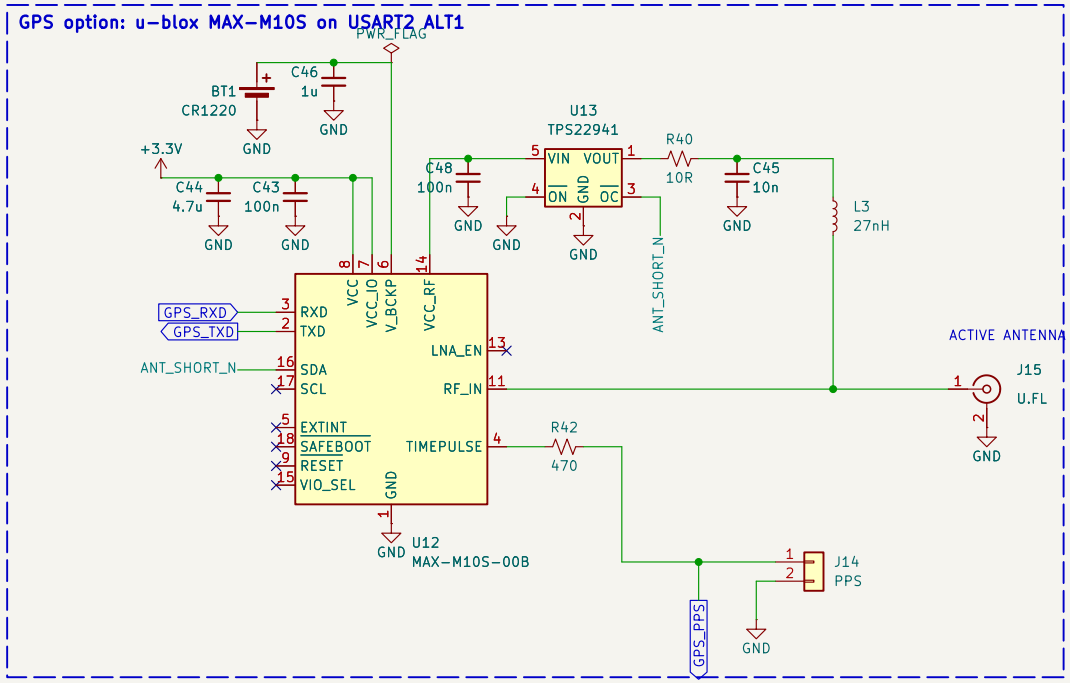

Boost Converter
IO Headers
Digital IO
ADC Inputs
I2C Bus
UART
User Interface
External EEPROM
Keying
Power Amplifier
PA Bias
Low-Pass Filter Bank
(sheet 2)
I2C Level Conversion
SI5351A PLL
MCU + USB
GPS
Filter Bank
File: filter_bank.kicad_sch
File: mcu_usb.kicad_sch
File: gps.kicad_sch
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Etherkit
Sheet: /
File: OpenBeacon2.kicad_sch
Title: OpenBeacon 2
Size: USLetter
Date: 2019-09-18
Rev: B
KiCad E.D.A. 10.0.7
Id: 1/4


F2 80/60 m low-pass (7th-order Cauer)
C202 220p
C204 330p
C206 120p
L201
L202
L203
F80_60_IN
F80_60_OUT
C201 360p
1.5u
C203 750p
1.5u
C205 820p
1.8u
C207 510p
GND
F3 40/30 m low-pass (7th-order Cauer)
C302 82p
C304 200p
C306 68p
L301
L302
L303
F40_30_IN
F40_30_OUT
C301 68p
680n
C303 430p
680n
C305 470p
1.0u
C307 360p
F4 20/17 m low-pass (7th-order Cauer)
C402 56p
C404 20p
C406 110p
L401
L402
L403
F20_17_IN
F20_17_OUT
C401 150p
390n
C403 330p
560n
C405 270p
270n
C407 1.1p
F5 15/12/10 m low-pass (7th-order Cauer)
C502 82p
C504 39p
C506 6.2p
L501
L502
L503
F15_10_IN
F15_10_OUT
C501 47p
180n
C503 160p
270n
C505 200p
330n
C507 110p
Relay tree (2 levels, input + output mirrored; all off = 80/60 m)
K1 = K_A1
pair sel. (in)
K1
G6K-2F-Y-DC5
P2_IN
P1_IN
LPF_IN
+5V
COIL_A
D1
1N4148W
K2 = K_B1
80/60 | 40/30 (in)
K2
G6K-2F-Y-DC5
F40_30_IN
F80_60_IN
P1_IN
COIL_B
D2
1N4148W
K3 = K_B2
20/17 | 15/10 (in)
K3
G6K-2F-Y-DC5
F15_10_IN
F20_17_IN
P2_IN
COIL_B
D3
1N4148W
K4 = K_A2
pair sel. (out)
K4
G6K-2F-Y-DC5
P2_OUT
P1_OUT
LPF_OUT
COIL_A
D4
1N4148W
K5 = K_B3
80/60 | 40/30 (out)
K5
G6K-2F-Y-DC5
F40_30_OUT
F80_60_OUT
P1_OUT
COIL_B
D5
1N4148W
K6 = K_B4
20/17 | 15/10 (out)
K6
G6K-2F-Y-DC5
F15_10_OUT
F20_17_OUT
P2_OUT
COIL_B
D6
1N4148W
Relay drivers
BAND_A
BAND_B
R9 22k
R10 22k
Q6 AO3400A
Q7 AO3400A
COIL_A
COIL_B
GND
BAND_A -> K1/K4, BAND_B -> K2/K3/K5/K6.
Q6 (BAND_A): 2 coils, 42 mA nom (5 V / 237 R),
<= 49 mA at 5.25 V, -10 % coil R.
Q7 (BAND_B): 4 coils, 84 mA nom, <= 98 mA.
AO3400A: 5.7 A cont., 48 mR @ Vgs 2.5 V -> <5 mV drop.
One 1N4148W flyback per coil, at each relay.
Band code + notes
Band code (BAND_A, BAND_B):
0 0 80/60 m (all relays off, NC path, default)
0 1 40/30 m (K2, K3, K5, K6 on)
1 0 20/17 m (K1, K4 on)
1 1 15/12/10 m (all six on)
Unselected filters float (open at both ends).
+5V relay load: 0 / 84 / 42 / 127 mA nominal (<=147 mA).
Relay pins: COM right, NC lower left, NO upper left; pole 2 unused.
Input and output paths never share a relay body.
Layout: keep K1-K2-K3 and K4-K5-K6 traces short,
50 ohm CPWG over solid GND, filters next to their relays.
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4 7th-order Cauer LPFs, 6-relay 2-level tree (2-bit BAND_A/BAND_B); unselected filters open
Etherkit
Sheet: /Filter Bank/
File: filter_bank.kicad_sch
Title: OpenBeacon 2 – Low-Pass Filter Bank
Size: A4
Date: 2026-10-25
Rev: C
KiCad E.D.A. 10.0.7
Id: 2/4


USB-C input + ESD
J12
USB_C_Receptacle
VBUS
CC1
CC2
D-
D-
D+
D+
SHIELD
GND
SBU1
SBU2
PWR_FLAG
+5V
R30
5.1k
R31
5.1k
USB_DM
USB_DP
D9
SP0503BAHT
USB-UART (CP2102N, from Si5351AVFO)
CP_VDD
R32
1k
C31
4.7u
C30
100n
C32
100n
C33
4.7u
R33
22.1k
R34
47.5k
VBUS
D-
D+
RST
VREGIN
VDD
VIO
TXD
RXD
RTS
CTS
DSR
DTR
DCD
RI/CLK
SUSPEND
TXT/GPIO.0
RXT/GPIO.1
RS485/GPIO.2
WAKEUP/GPIO.3
MCU_RX
MCU_TX
DTR
U10
CP2102N-A02-GQFN24
+3.3 V regulator (TLV75533, 500 mA)
U11
C34
1u
C36
4.7u
IN
EN
OUT
NC
TLV75533PDBV
TLV755P DS (Table 4-1, ROC): CIN, COUT >= 1 uF ceramic (>= 0.47 uF effective),
COUT <= 200 uF. EN tied to IN. Dropout <= 238 mV @ 500 mA.
IQ 25 uA, PSRR 46 dB @ 100 kHz. Pin 4 NC (LP2985 BP cap C35 removed).
UPDI header + DTR auto-reset (from Si5351AVFO)
J13 copied from Si5351AVFO J5
(EtherKeyerMini pinout):
1 VCC ref only, 2 TXD,
3 RX/UPDI, 4 GND.
No solder jumpers.
J13
UPDI_TXD
UPDI_RX
UPDI
D10
1N5711W
R35
470
R37
10k
C37
100n
R36
1k
RESET
MCU (AVR128DA32, DxCore, 3.3 V)
U9
C39
1u
C38
100n
C40
100n
C41
15p
C42
15p
32.768kHz
UPDI
RESET/PF6
XTAL32K1/PF0
XTAL32K2/PF1
VDD
AVDD
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PD0
PD1
PD2
PD3
PD4
PD5
PD6
VREFA/PD7
PC0
PC1
PF5
PC3
PF2
PF3
PF4
PC2
SDA
SCL
TX
BAND_A
BAND_B
D4
UART_TX
UART_RX
GPS_TXD
D10
BTN_DSP_1
BTN_DSP_2
GPS_RXD
BTN_UP
BTN_DOWN
BTN_LEFT
BTN_RIGHT
BTN_BACK
AVR128DA32-I/PT
R38
470
D11
SYNC
R39
470
D12
TX
VERIFY
[ ] U9 pin map: TWI0 PA2/PA3, USART0 PA0/PA1 (CP2102N), USART1 PC0/PC1 (J10), ADC PD0-PD2 (A2-A4),
BAND_A PA5 / BAND_B PA6, D4 PA7, SYNC LED PC2, GPS USART2 ALT1 PF4 TX / PF5 RX; firmware pin table must follow it.
[ ] GPS_PPS MCU pin undecided (A: PC2 + drop D4 or D10, B: PC3 instead of D10, C: one of PD0-PD2).
[ ] Y1: CL 12.5 pF, C41/C42 15p assumes 5 pF pin + ~2 pF stray (XOSC32K LPMODE=0);
trim after measuring the 32 kHz error.
[x] TX LED D12 (KT-0603YG) + R39 470R hang off the TX net (PA4): ~3 mA, no extra MCU pin.
[x] D10/R35 UPDI diode + 470R identical to Si5351AVFO J5 and EtherKeyerMini J3.
[ ] Q6/Q7 AO3400A drive the 6-relay tree (Q6 2 coils ~42 mA, Q7 4 coils ~84 mA); Q8/Q9 removed.
[ ] +5V now from USB-C VBUS only (500 mA default USB); PA draw not yet measured.
[x] U1 FB tied to VIN: Jason: FB tied to VIN selects internal 12 V output (TLV61046A datasheet).
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AVR128DA32 (DxCore) replaces Empyrean SAMD21; CP2102N/UPDI from Si5351AVFO
Etherkit
Sheet: /MCU + USB/
File: mcu_usb.kicad_sch
Title: OpenBeacon 2 - MCU, USB-UART and 3.3 V
Size: A4
Date: 2026-10-25
Rev: C
KiCad E.D.A. 10.0.7
Id: 3/4



VERIFY

- [x] Antenna short: TPS22941 limit 40–80 mA, auto-restart, avg <= 27 mA (was 320 mA).
- [x] V_BCKP: CR1220 in KH-BS1220-2-SMT (C5365932): C47/D13/R41 removed.
- [ ] BT1 footprint: draw from KH-BS1220-2-SMT DS.
- [ ] U13 TPS22941 JLC stock thin (26 pcs).
- [ ] Short peak 80 mA > VCC_RF 50 mA op. (DS T13) for <= 20 ms bursts; confirm on bench.
- [ ] V_IO ramp 25–35000 us/V (DS 4.1): measure.
- [ ] GPS_PPS MCU pin: see mcu_usb.

RECEIVER: VCC + V_IO tied to +3.3V; VIO_SEL open = 3.3 V I/O (IM 4.1.2).
C43/C44 beside pins 7/8, short GND vias; no series R > 0.2 ohm in VCC (IM 4.1.1).
RESET_N, EXTINT, SCL, SAFEBOOT_N, LNA_EN left open (DS Table 10).
SDA = ANT_SHORT_N (IM 3.3.1.2): CFG-HW-ANT_CFG_SHORTDET=1, ANT_SUP_SHORT_PIN=SDA PIO.
PPS: TIMEPULSE = SAFEBOOT_N via 1k inside (DS fn 15); nothing may hold it low at power-up, so no pull-down. R42 limits a shorted J14 to ~7 mA (I_PIO 10 mA).

ANTENNA: RF_IN is DC-blocked inside the module (IM Table 1), no external DC block.
Bias-T VCC_RF -> U13 TPS22941 -> R40 10R -> C45 10n -> L3 27 nH onto RF line (IM 3.3.1.2, 4.3.4).
U13: I_LIM 40–80 mA, 5–20 ms blanking, 40–160 ms restart (TPS2294x DS 7.5): short avg <= 27 mA, typ 7 mA (VCC_RF abs max 250 mA, DS Table 12). 25 mA antenna drop ~0.27 V -> ~2.9 V at J15.
Layout: 50 ohm CPWG pin 11 -> J15, L3 pad on the line, >= 5 mm from digital.
BACKUP: BT1 CR1220 (3 V, 35 mAh, user-fitted, NOT rechargeable, no charger) on V_BCKP;
C46 1u at pin 6 (IM 4.1.3). 35 mAh / 28 uA (DS Table 17) = ~1250 h (~52 days) powered off.

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MAX-M10S on USART2 ALT1, PPS to J14, current-limited active-antenna bias-T, CR1220 V_BCKP

Etherkit

Sheet: /GPS/
File: gps.kicad_sch

Title: OpenBeacon 2 – GPS (u-blox MAX-M10S)

Size: A4	Date: 2026-10-25	**Rev: C**
KiCad E.D.A. 10.0.7		Id: 4/4